Gen AI Architect Program

Assignment 7 - AI Automation: ETL Pipeline Challenge | learn. build. operate

1. Problem statement

You are working as a Data Engineer at **Social Eagle AI Academy**. The academy receives raw student enrollment data every week from Google Forms. The data is messy, inconsistent, and full of errors. Your job is to build a complete **ETL AI automation pipeline** using **n8n** that extracts the raw student data from a CSV file, uses **AI to clean** every row automatically, applies **two conditions** to filter the data, and saves the final result as clean files.

Difficulty: Medium · Tools: n8n, an LLM, CSV / Google Sheets

2. Your task

Build an n8n workflow that extracts the data, cleans each row with AI, and routes every row through the two conditions below.

Condition 1 - Location filter

```
If City = Chennai -> CHENNAI sheet / file
If City = Other  -> OTHER CITIES sheet / file
```

Condition 2 - Email filter

```
If Email is VALID -> VALID EMAIL sheet / file
If Email is INVALID -> INVALID EMAIL sheet / file
```

3. Dataset

Use the same file from class - `student_enrollment_raw.csv` - which contains 50 rows of messy, real-world student data.

4. Expected output

Combining both conditions gives four output files / sheets:

Output file / sheet	Condition
`chennai_valid_email`	City = Chennai AND Email valid
`chennai_invalid_email`	City = Chennai AND Email invalid
`other_valid_email`	City = Other AND Email valid
`other_invalid_email`	City = Other AND Email invalid

Four output sheets in total.

5. Your n8n workflow must have these nodes

Node	Purpose
Form Trigger or Manual Trigger	Start point
Extract from File	Read the CSV
Loop Over Items	Process row by row
Basic LLM Chain	AI cleans each row
Code Node	Parse the AI response
IF Node 1	Split by City
IF Node 2	Split by Email validity
Google Sheets x4 or Convert to File	Save the outputs

6. AI cleaning rules (use this prompt)

Use this exact prompt in your Basic LLM Chain node:

```
Clean this student enrollment row and return ONLY a JSON object.
Row data: {{ JSON.stringify($json) }}

Rules:
- Name: Title Case
- Email: if invalid (no @ or no .com/.in) set INVALID_EMAIL
- Phone: if less than 10 digits or empty set MISSING
- Course: only Python or Machine Learning or Data Science
- Fee_Paid: yes/YES = true, no/NO = false
- City: Title Case, empty = UNKNOWN
- Enrolled_Date: YYYY-MM-DD format

Return ONLY raw JSON. No explanation. No markdown.
```

7. Bonus challenge

If you finish early - add a 5th output

Students from Chennai with a valid email AND Fee_Paid = true. Save them separately as chennai_paid_confirmed.

8. Submission checklist

Before you submit, check that:

[ ] Workflow runs without error

[ ] All 50 rows processed by AI

[ ] City condition working correctly

[ ] Email condition working correctly

[ ] 4 output sheets created with correct data

[ ] Screenshot of the final n8n canvas

[ ] Screenshot of all 4 output sheets

9. What to submit

- Export your n8n workflow (JSON) and push it to **GitHub**.
- Share your GitHub repository link in **WhatsApp**.
- Share your n8n canvas screenshot in the **WhatsApp** group.
- Post your screen-recorded video of the workflow running in the **community**.

Deadline: submit before the next class session.

All the best! Build it, break it, fix it - that is how you learn. Build the workflow yourself; the spec above is your guide, not the finished answer.